

Practice Quiz: Cognition — Hidden Layers in Hook and Mind

Part A: Multiple Choice

1. In the crochet-as-neural-network analogy, the foundation chain corresponds to: A) The output layer — it is the final result B) The input layer — it is the raw starting data upon which all hidden layers build C) A hidden layer — it contains emergent textures D) The loss function — it measures how wrong the project is
2. A shell stitch (five double crochets into one stitch) is best described as: A) An input feature — it is the raw data entering the system B) A hidden representation — an emergent texture that is not present in any single double crochet alone C) An output label — the final classification of the fabric D) A bias term — a constant added to every layer
3. The crocheter's multi-level mental model (stitch, row, section, project, design) is an example of: A) A flat, single-layer representation with no hierarchy B) A deep generative model with multiple levels generating predictions about the levels below C) A purely bottom-up process with no top-down predictions D) A lookup table that contains every possible crochet project
4. When a crocheter frogs several rows to fix an error, the cognitive cost is high because: A) Yarn is expensive B) The emergent textures (hidden representations) built over those rows are destroyed and must be rebuilt, and the crocheter must reposition their mental model in the hierarchy C) Frogging requires no thought at all D) The foundation chain is always destroyed when frogging
5. An experienced crocheter can improvise (work without a pattern) because: A) They do not need a generative model — they work randomly B) Their deep generative model generates precise predictions at every level, from finished object down to individual stitch C) They have memorized every possible crochet pattern D) Improvisation does not involve cognition

6. Stitch markers function as: A) Decorations with no cognitive purpose B) External memory — physical tokens that carry information, offloading cognitive load from the crocheter's internal model C) Replacements for the entire generative model D) Input layer elements that disappear in the first hidden layer
7. In Active Inference, top-down predictions in the crocheter's deep generative model means: A) The crocheter only processes information from the current stitch, never considering the overall project B) The crocheter's vision of the finished object generates expectations about what each section, row, and stitch should look and feel like C) Information only flows from bottom to top, never from top to bottom D) The crocheter must always look at the project from above

Part B: Short Answer

1. Choose a multi-row stitch pattern (such as a cable, a chevron/zigzag, or a wave pattern). Explain how this pattern creates an emergent hidden representation that spans multiple "layers" of the fabric. Why can you not see the pattern by looking at a single row in isolation?
2. A crocheter picks up a half-finished project after three weeks and cannot remember where they left off. Describe this situation using the concept of a deep generative model that has lost its state. What information has been lost at each level of the hierarchy (stitch, row, section, project), and what strategies might the crocheter use to reconstruct it?
3. Compare a beginner following a written pattern step by step to a neural network performing inference with pre-trained weights. Then compare an expert improvising without a pattern to a generative model producing novel outputs. What is the critical difference in the depth and richness of the internal model between the two crocheters, and how does this parallel the difference between a shallow and a deep neural network?